Category 1: Artificial Intelligence & Prompt Engineering

1. What is an LPU and why use Groq over OpenAI/GPT-4?

- **Answer:** Groq uses LPUs (Language Processing Units), hardware specifically designed for deterministic AI inference, unlike general-purpose GPUs. This gives our application near-instantaneous response times, which is critical for a real-time developer tool.

2. What happens if the AI ignores your prompt and doesn't output the |||SPLIT||| delimiter?

- **Answer:** Our backend handles this edge case. If text.split("|||SPLIT|||") returns an array with fewer than 3 elements, the system defaults the status to "WRONG" and passes a fallback error message ("Explanation format error from AI") so the app doesn't crash.

3. If I paste an essay instead of code, what will your AI do?

- **Answer:** Because of the system prompt constraints, it will likely evaluate it as "WRONG" (invalid syntax) and the explanation will state that the provided text is not valid code.

Category 2: Backend & Python (Flask)

4. Why did you choose Flask instead of Django or FastAPI?

- **Answer:** Flask is a micro-framework. Since our backend primarily acts as a routing controller for the Groq API and MongoDB, we didn't need the heavy ORM and built-in features of Django. It kept our backend lightweight and fast.

5. How are you managing user sessions?

- **Answer:** We use Flask's built-in session object, which stores the user_id and username in a cryptographically signed cookie on the client side. This ensures users can only see their own dashboard history.

6. What is load_dotenv() doing in your app.py?

- **Answer:** It loads sensitive environment variables, like the MongoDB URI and secret keys, from a .env file that is *not* pushed to GitHub. This prevents credential leaks.

6. How does your backend handle markdown backticks (```) coming from the AI?

- **Answer:** We wrote a custom Python cleanup block that checks if the string starts and ends with ```. It splits the string by newlines, removes the first and last lines, and joins it back together, ensuring the frontend gets pure, executable code.

Category 3: Database (MongoDB)

7. Why NoSQL (MongoDB) instead of SQL (MySQL)?

- **Answer:** Code snippets and explanations vary wildly in length and structure. A document-based NoSQL database allows us to store these dynamic payloads as BSON documents without worrying about strict character limits or rigid table schemas required by SQL.

8. What is the ServerSelectionTimeoutError in your code?

- **Answer:** It is an exception thrown by PyMongo if the Flask server cannot connect to the MongoDB Atlas cloud cluster (e.g., due to internet issues or IP whitelisting). We catch this to prevent the app from crashing and display a flash message to the user.

9. Explain how generate_password_hash makes your app secure.

- **Answer:** It uses the PBKDF2 algorithm with a SHA-256 hash and a unique salt. Even if two users have the password "1234", their database hashes will look completely different, preventing rainbow table attacks.

10. Why did you store user_id as a string in the session instead of an ObjectId?

- **Answer:** Flask sessions serialize data into JSON to store in cookies. MongoDB's native ObjectId is not JSON serializable, so it must be cast to a string first.

Category 4: Frontend (JS, HTML, CSS)

11. How does your JavaScript handle communication with the Python backend?

- **Answer:** We use the modern ES6 fetch API. It sends an asynchronous HTTP POST request containing a JSON payload (the code and toggle states). We await the response so the UI doesn't freeze during processing.

12. Why are you using marked.js?

- **Answer:** The AI returns explanations formatted in Markdown (using asterisks for bolding, backticks for code). Browsers only understand HTML. marked.js safely parses that Markdown string into renderable HTML for the DOM.

13. What is Prism.highlightElement(codeDisplay) doing?

- **Answer:** After the fixed code is injected into the DOM, it tells the PrismJS library to scan that specific HTML block and apply lexical tokenization to color-code the syntax (keywords, strings, functions).